

Q6) Logistic Regression

⑥

$$P(y=1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

$$\beta_0 = -3$$

$$\beta_1 = 0.6$$

$$x=3$$

$$= \frac{1}{1 + e^{-(3 + 0.6 \cdot 3)}}$$

$$= \frac{1}{1 + e^{-1.2}}$$

$$x=3$$

$$= \frac{1}{4.91} = 0.2031$$

$$P = 20.31\%$$

⑧

$$x=8$$

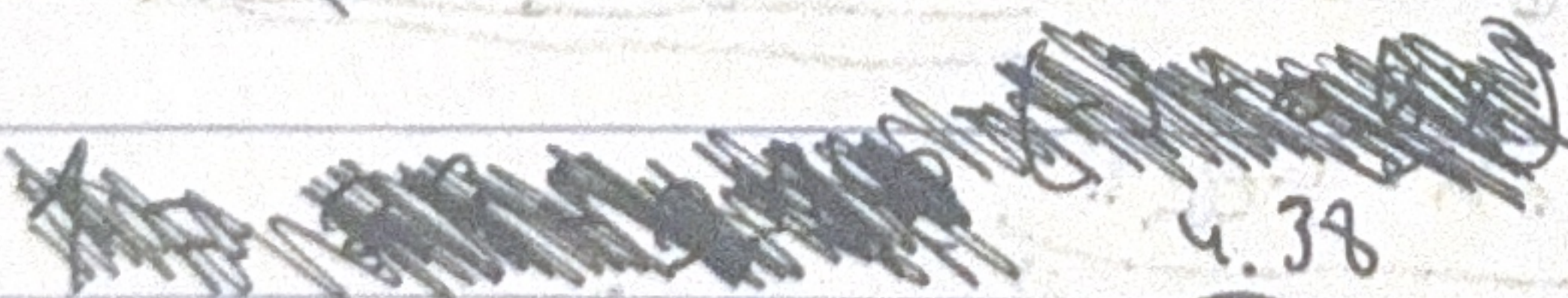
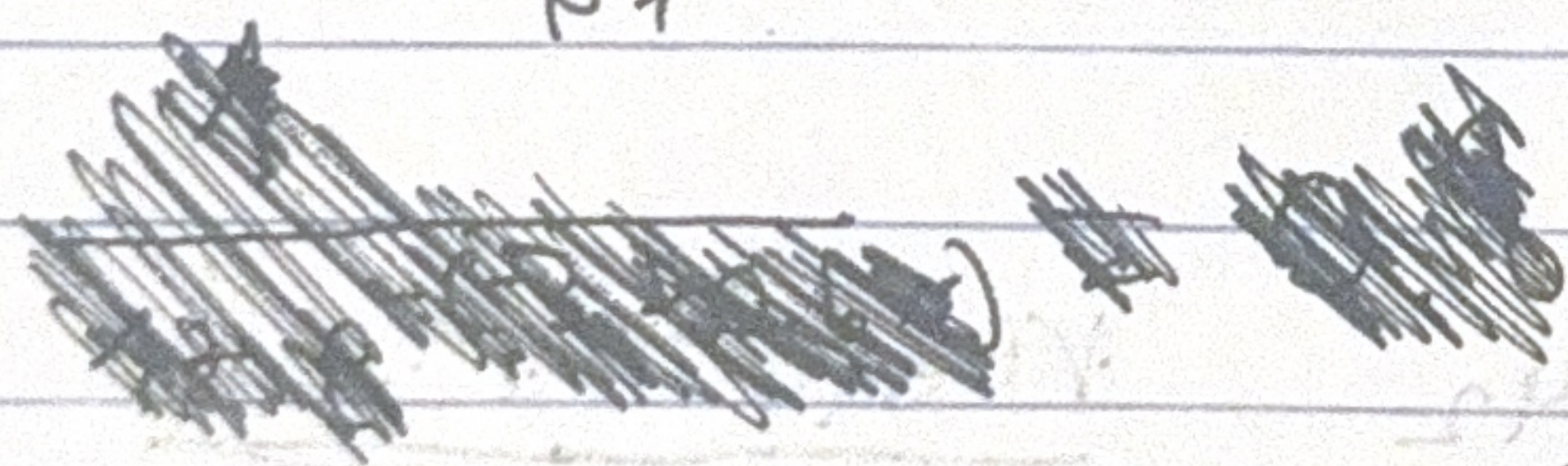
$$= \frac{1}{1 + e^{-(3 + 0.6 \cdot 8)}} = \frac{1}{1 + e^{-1.8}}$$

$$= \frac{1}{1.165} = 0.859$$

$$P = 85.9\%$$

⑨

$$x(p) = \frac{\ln\left(\frac{p}{1-p}\right) - \beta_0}{\beta_1}$$



$$= \frac{\ln\left(\frac{30}{20}\right) + 3}{0.6}$$

$$= \frac{1.38 + 3}{0.6}$$

$$= 7.3$$

★ Bonus ★

the threshold for 80% is 7.3 suspicious keywords